Visual Investigator: Running & Deployment Guide

How to run the app on Windows, stop and restart it, push it to GitHub, and deploy it free on Streamlit Community Cloud.

1. Run the app on your computer

Open **PowerShell** and run these commands in order. Steps marked *first time only* are needed just once per computer.

Step 1. Go to the project folder:

```
cd E:\AI-Agents\visual_investigator
```

Step 2 (first time only). Create a virtual environment:

```
python -m venv .venv
```

Step 3 (first time only). Install the required packages:

```
.venv\Scripts\python -m pip install -r requirements.txt
```

Step 4 (first time only). Create the **.env** file, then open it and paste your API key after `ANTHROPIC_API_KEY=`. Never put the key in .env.example.

```
copy .env.example .env
```

Step 5. Start the app. It opens at **http://localhost:8501**:

```
.venv\Scripts\python -m streamlit run app.py
```

Optional. Analyse the whole dataset from the terminal instead of the UI:

```
.venv\Scripts\python batch.py
```

Optional. Run the offline tests (no API key or internet needed):

```
.venv\Scripts\python tests\test_offline.py
```

Note: `ui_preview.html` is only a static design mock-up with sample data. It cannot analyse images. Always use the Streamlit app above.

2. Stop and start the app

Action	How
Stop	Click the PowerShell window running the app and press **Ctrl + C** (closing that window also stops it).
Start again	Run Step 1 (`cd ...`) and Step 5 (`streamlit run`) again.
Port busy	If port 8501 is already used, Streamlit opens on 8502 instead. Read the URL printed in the terminal.

3. Put the project on GitHub

GitHub stores your code. It cannot run a Python/Streamlit app by itself (GitHub Pages only hosts static pages). So you push the code to GitHub, then Streamlit Community Cloud runs it from there (section 4).

Step 1. On **github.com** click **+ > New repository**. Name it, e.g. `visual-investigator`. Choose Public or Private. Do **not** tick "Add a README" (the project already has one). Click **Create repository**.

Step 2. In PowerShell, inside the project folder, check who commits will be credited to (change it if it is not you):

```
git config --global user.name
git config --global user.email

git config --global user.name "Your Name"
git config --global user.email "you@example.com"
```

Step 3. Create the local repository and stage the files:

```
git init
git add .
```

Step 4. Safety check. List the staged files and make sure **.env** and **.venv** are NOT in the list (.gitignore already excludes them):

```
git status
```

Step 5. Commit, then connect to GitHub and push. Replace `YOUR-USERNAME` with your GitHub username:

```
git commit -m "Visual Investigation Agent"
git branch -M main
git remote add origin https://github.com/YOUR-USERNAME/visual-investigator.git
git push -u origin main
```

On the first push a browser window asks you to sign in to GitHub. After that, refresh the repository page and your files appear.

Later updates. After changing code, push again with:

```
git add .
git commit -m "Describe the change"
git push
```

4. Deploy free on Streamlit Community Cloud

Step 1. Go to **share.streamlit.io** and sign in with your GitHub account.

Step 2. Click **Create app** and choose **Deploy a public app from GitHub**.

Step 3. Pick your repository, branch `main`, and set **Main file path** to `app.py`. Optionally choose a custom URL.

Step 4. Open **Advanced settings**, choose Python 3.13 (or 3.12), and in **Secrets** paste:

```
ANTHROPIC_API_KEY = "sk-ant-your-new-key"
```

Step 5. Click **Deploy**. The first build takes a few minutes. You then get a free public link like `https://your-app.streamlit.app`. Every `git push` redeploys it automatically.

Alternative free host: Hugging Face Spaces (Docker/Streamlit template). Streamlit Cloud is simpler for this project.

Important before you deploy

• **Revoke the old API key.** It was stored in .env.example, which is meant to be shared. Create a new key in console.anthropic.com and put it only in .env (local) and Streamlit Secrets (cloud).

• **Hosting is free, the Claude API is not.** Everyone using your public link spends your API credit. Set a monthly spend limit in the Anthropic Console.

• **Shared-key issue:** a key pasted in the sidebar is currently stored for the whole server, so on a public deployment it would be used for every visitor. Fix this before sharing the link widely, or rely only on Secrets.

• On Streamlit Cloud, uploaded images and new reports are lost when the app restarts. The reports in outputs/ that you commit to GitHub stay available.